Course Outline Template

Indian Institute of Management Calcutta

Programme: MBA

Course Code:

Name of the Course: Decoding the Black-Box: Responsible AI & Future Tech Strategies

Core / Elective: Elective **Cap (for Elective)**:60

Credit: 3

Academic Year & Term: 2025 – 26, Term VI

Instructor: Rahul Kumar

Affiliation: MIS Group, IIM Calcutta

Email: rahulk@iimcal.ac.in

Introduction to the Course:

In today's world, technological advancements are unfolding at an unprecedented rate. Emerging technologies, once nascent, are now central to significant disruptions that impact both business and society, at large. As these technologies converge, they introduce innovative norms and solutions that challenge existing work practices. This rapidly evolving technological landscape presents its own set of challenges, particularly for the future of work. Navigating this era of disruption effectively, and leveraging these advancements to one's advantage, remains a critical concern.

Artificial Intelligence (AI) has become a pivotal force in this transformative environment. Its influence extends beyond its own domain, significantly accelerating the development of other technologies such as the Metaverse, Immersive Platforms, Cloud Computing, and Content Creation through Generative AI. The impact of AI on these fields is profound and unparalleled, necessitating that managers and future leaders recognize and adapt to these disruptive changes.

These technologies have the potential to replace traditional practices, offering novel solutions and altering the status quo.

This course aims to provide participants with a fresh perspective on the responsible management of these disruptive forces shaping the future of work. It adopts a fundamental approach to understanding AI, emphasizing its potential to transform work practices. Additionally, the course advocates for the democratization of AI, ensuring equitable access and broad participation in technologies such as Generative Models and Automated Content Creation. By exploring the integration of AI with other technological advancements, this module seeks to uncover the modern strategies and components that will influence the future of work. The course asserts that a cohesive understanding of technologies and how their integration with each other can act as precipitators for offering best practices. Summarily, by examining how AI integrates with other emerging technologies, the course aims to equip managers with the knowledge needed to adeptly navigate and shape the future of work.

Course Objectives and Intended Outcomes:

In order to explore the evolving landscape of the future of work, it is essential to understand how new technologies are driving innovative practices. It therefore requires a growing need for specialized skills, especially in IT and digital fields. The course thus intends to foster:

- Knowledge about explainable AI/ML models and related best practices
- Understanding about aspects of transparency, fairness, bias and flexibility of models
- Awareness about the importance of equality and ubiquity of AI
- Knowledge about cloud computing, immersive technologies, metaverse and Generative AI
- Practical insights about integrating and managing these novel technologies
- Envisioning implications of futuristic technologies on the future of work

Mapping with Program Level Learning Goals:

This course is majorly aligned with two program level goals, viz., critical thinking and technological competency. Specifically, after completing the course, students would be able to:

Goal 1 - Critical Thinking	a	Critically think and analyse resolution for important contemporary problems and issues
	c	Use relevant tools and models to analyse situations and solve practical business problems

	d	Acquire the ability to recommend managerial solutions from the process of decision analysis
Goal 5- Technological Competencies	a	Learn analytical tools, techniques and methods for solving problems
	c	Use appropriate analytical orientation to identify and assess digital artifacts for competitive advantage

International Component in your course (if any):
Since the course pivots around futuristic technologies, most of the applications and discussions would pertain to International Settings and Standards. Nevertheless, the discussions and applications would also be linked to the Indian contexts.

Pre-requisites:

The participants should be able to apply their knowledge from 'Statistics for Management' for drawing conclusions. Takeaways from the second module (post mid-term) of 'Information Technology & Systems' may further be helpful.

Recommended Book:
Responsible AI: Implementing Ethical and Unbiased Algorithms, Springer (2021).

Tentative Session Plan:

THEME	SESSION NO	TOPIC	Recommended Readings
Responsible AI	1	➢ When AI Misbehaves ➢ The Canvas of Responsible AI ➢ The Trap of Unfavourable Outcomes	https://hbr.org/2022/08/when-and-why-you-should-explain-how-your-ai-works?ab=hero-main-text Chapter 1 of Recommended Book- Introduction-**Facets of responsible AI**
The Precursors	2	➢ Is Data Quality all about Hygiene ➢ Data Quality as the Beginning Point ➢ Traceability and Single Source of Truth	Janssen, M., Brous, P., Estevez, E., Barbosa, L. S., & Janowski, T. (2020). Data governance: Organizing data for trustworthy Artificial Intelligence. *Government information quarterly*, *37*(3), 101493.

			Mark Chun, Owen Hall, Charla Griffy-Brown (2016), Cloud Syzygy Technologies: Cloud-Based Computing, Ivey Publishing, September 6, 2016 *Industry talk
Explainable AI	3	➢ The Importance of Transparency and Explainability ➢ The Taxonomy of Interpretable/Explainable Models 'by Design'	Chapter 4 of Recommended Book- **Explainability**
Flexibility of AI Models	4	➢ Understanding Bias ➢ Foundations of Bias-Variance Trade-off ➢ The Guidelines of Principle of Parsimony in algorithm development	Chapter 3 of Recommended Book- **Bias in Data**
Fairness of AI models	5	➢ Mathematical Transformations for Feature Engineering ➢ Curating Proxy Features ➢ Decoding the black box nature of AI models	Chapter 2 of Recommended Book- **Fairness and Proxy Features**
Model Accountability	6	➢ Variable Selection Strategies ➢ Tools, Techniques and Frameworks ➢ The Power of Simple Models	COMPAS Case Study: Investigating Algorithmic Fairness of Predictive Policing | by Mallika Chawla | Medium
Model Reliability	7	➢ The concepts of Elastic nets ➢ Balancing Complexity and Performance through Regularization ➢ Shrinkage Methods to undermine noisy factors	Yeh,I-Cheng. (2007). Concrete Compressive Strength. UCI Machine Learning Repository. https://doi.org/10.24432/C5PK67. Solving a Kaggle Competition: https://www.kaggle.com/datasets/elikplim/concrete-compressive-strength-data-set

Post-Hoc Explanations	8	➢ The concept of Information ➢ Decoding the significance of information theory ➢ Information gain, Randomness and Entropy	Equality and fairness measures in classification models | Auditing machine learning algorithms (auditingalgorithms.net) A gentle introduction to ML fairness metrics | Superwise ML Observability
Can Trust be Objective?	9	➢ Methods for Calibrating Decision Boundary ➢ Differences and similarities between log entropy and cross-entropy ➢ Equalizing Odds	Chapter 5 of Recommended Book- **Removing Bias from ML Model**
Equitable Evaluation	10	➢ How Imbalance leads to Inequality ➢ Addressing Evaluation Bias ➢ The Perils of Accuracy	Chapter 6 of Recommended Book- **Removing Bias from ML Output**
AI Accessibility and Inclusivity	11	➢ Ensuring AI tools are accessible to diverse user groups ➢ Designing AI systems for underrepresented communities ➢ Strategies for inclusive data collection and model development	https://hbr.org/2022/06/building-transparency-into-ai-projects Marianne Kinnula, Netta Iivari, et al.,. 2021. Researchers' Toolbox for the Future: **Understanding and Designing Accessible and Inclusive Artificial Intelligence (AIAI**). In Proceedings of the 24th International Academic Mindtrek Conference (Academic Mindtrek '21). Association for Computing Machinery, New York, NY, USA, 1–4. https://doi.org/10.1145/3464327.3464965

Tech trait I: Democratization of AI	12	➢ Democratization of Development ➢ The role of Responsible AI ➢ How Explainability and Transparency Influence Democratization	**Jio Promises AI To Everyone, Everywhere**:https://youtu.be/cwm9XhnRHKw?si=0aPrKycACNJup5do Greenstein, Shane, and Christine Snively. "Viacom: Democratization of Data Science." Harvard Business School Case 618-016, January 2018. *Industry talk
	13	➢ Democratization by Use ➢ Generative (Gen) AI at work ➢ For Anyone and Everyone	Turn Generative AI from an Existential Threat into a Competitive Advantage. Stanton, Christopher, and Matt Higgins. "Generative AI and the Future of Work." Harvard Business School Case 824-130, December 2023. (Revised February 2024)
Tech trait II: Technology Acceleration	14	➢ The Symbiosis of AI & Cloud ➢ Scaling Cloud Services with AI ➢ Global Reach and Accessibility	https://hbr.org/2011/11/what-every-ceo-needs-to-know-about-the-cloud Lai, Wen-Chi and Hung, Wei-Hsi, "A Framework of Cloud and AI based Intelligent Hotel" (2018). *ICEB 2018 Proceedings (Guilin, China)*. 99. https://aisel.aisnet.org/iceb2018/99

Tech trait III: Technology Immersion	15	➢ Serendipity In the Internet Age ➢ Contextually aware Immersive Experiences with Augmented Reality ➢ Exploring Virtual Reality for enhancing real-world experiences	Farshid, M., Paschen, J., Eriksson, T., & Kietzmann, J. (2018). Go boldly!: Explore augmented reality (AR), virtual reality (VR), and mixed reality (MR) for business. *Business horizons*, *61*(5), 657-663 Pradhan, Sandip and Chattopadhyay, Manojit (2024), Exide Industries Limited: Pioneering Remote-as-a-Service through Augmented Reality, Ivey Publishing, W35236, 24-02-02
Tech trait IV: Ecosystems	16	➢ Three tenets of Metaverse: Experience, Identity and Ownership ➢ The Human Centricity and Interaction ➢ Custom Metaverse Avatars	Fortnite ARIANA GRANDE Concert (NO COMMENTARY) Purdy, M. (2023). Building a great customer experience in the metaverse. *Harvard Business Review*, *3*. Lekushoff, A. (2022). To Metaverse or Not to Metaverse isn't the Question. *Ivey Business Journal Reprints*, 1-7.
	17	➢ Digital Economy and Trade within the Metaverse ➢ Recreating Experiences through Interoperability ➢ Inclusivity in the Metaverse	W. Chan Kim, Renee Mauborgne, Michael Olenick (2022), Meta: Facebook's Pivot to the Metaverse. A Path to Dystopia or Blue Ocean Utopia?, INSEAD, 19 January 2022. Baker, K. R. (2022). Designing an Inclusive Metaverse. *Harvard Business Review*. Dwivedi, Y.K., Kshetri, N., Hughes, L. *et al.* Exploring the Darkverse: A Multi-Perspective

			Analysis of the Negative Societal Impacts of the Metaverse. *Inf Syst Front* **25**, 2071–2114 (2023). https://doi.org/10.1007/s10796-023-10400-x
Tech trait V: Convergence of Technologies	18	➢ Replicating Physical Systems & Processes ➢ Digital Twins and The Phy-Gital world ➢ The Next Level of innovation	Apte, P. P., & Spanos, C. J. (2021). The digital twin opportunity. *MIT Sloan Management Review*, *63*(1), 15-17. Tozanli, Ö., & Saénz, M. J. (2022). Unlocking the potential of digital twins in supply chains. *MIT Sloan Management Review*, *63*(4), 1-4.
Navigating the Tech Traits	19	➢ The Nightmares of Emerging Technologies ➢ The Difficult Road of Governance ➢ Governance is not Just about Frameworks	https://hbr.org/2023/08/generative-ai-nxiety?ab=HP-hero-featured-text-1 https://hbr.org/2023/05/how-to-avoid-the-ethical-nightmares-of-emerging-technology Practitioner's Perspectives
Concluding Session	20	➢ The Road Ahead ➢ Ethical Considerations ➢ Society at Large	Agrawal, A., Gans, J., & Goldfarb, A. (2017). What to expect from artificial intelligence. Gans, J. S. (2016). Keep calm and manage disruption. *MIT Sloan Management Review*, *57*(3), 83. Gans, J. (2016). The other disruption. *Harvard Business Review*, *94*(3), 78-84. Practitioner's Perspective

Evaluation (TBD):

Quiz	Individual	30%
Assignment	Group	20%
White Paper Submissions	Group	50%